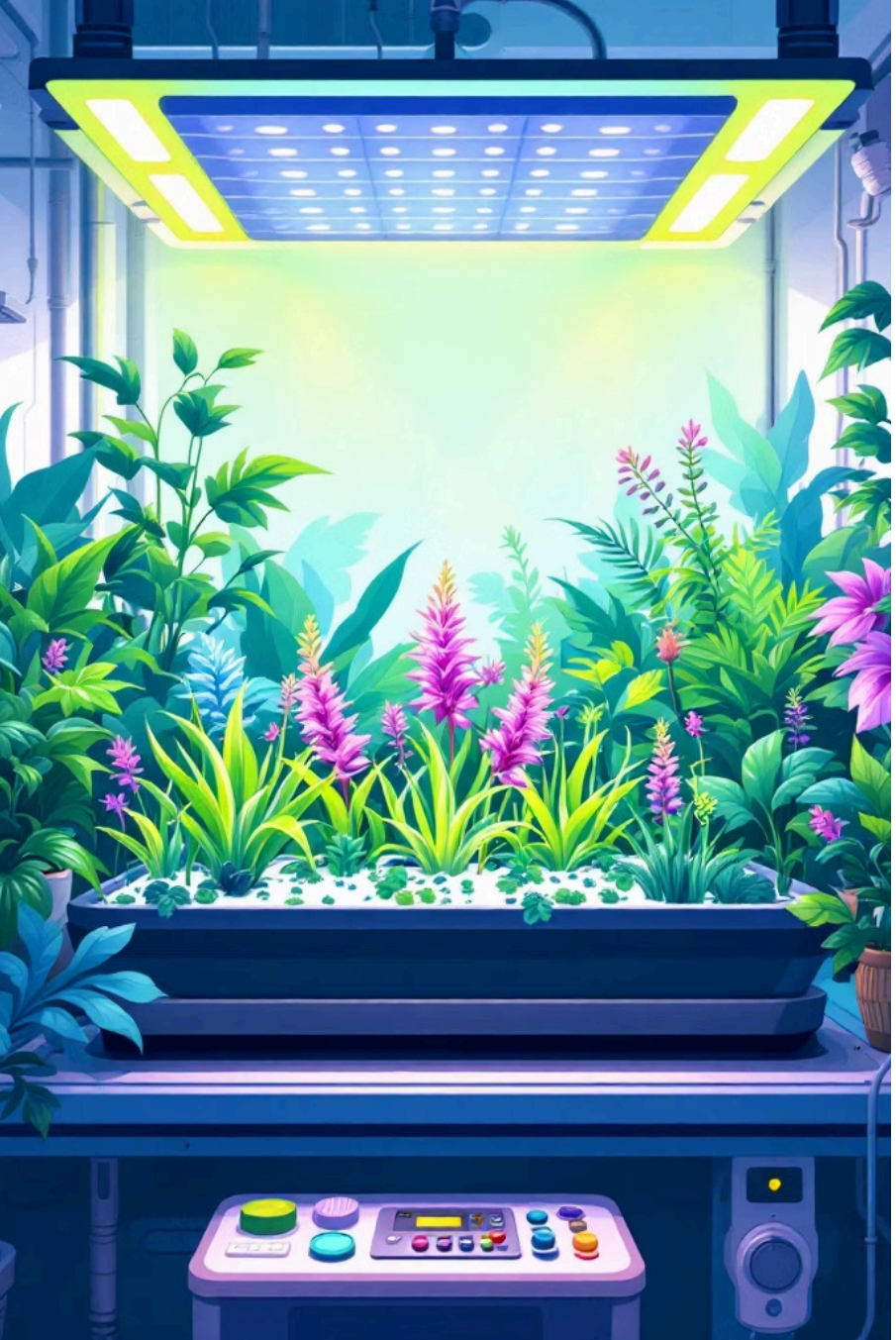

NutriGrow AI: Revolutionizing Exotic Plant Cultivation with Smart Hydroponics

An intelligent IoT and AI-driven platform for precision cultivation of rare and exotic plant species in controlled environments.

Made with GAMMA

Problem Statement: The Limits of Traditional Agriculture for Exotic Plants

Resource Intensive

Traditional farming demands vast land, water, and labor, often leading to unsustainable practices that strain environmental resources.

Environmental Vulnerability

Exotic plants with specific needs are highly susceptible to unpredictable climate, pests, and diseases in open environments.

Limited Scalability

Geographic constraints and climate dependency restrict the widespread cultivation and availability of rare plant species.

Quality Inconsistency

Manual monitoring leads to variations in nutrient levels, pH, and environmental conditions, impacting plant health and market quality.

Made with GAMMA

Methodology: A Phased Approach to Smart Cultivation

System Design & Hardware

Develop closed-loop hydroponic system with integrated sensors and actuators controlled by ESP32 microcontroller.

Data Acquisition

Collect real-time sensor data and high-resolution plant imagery. Clean and format data for ML training.

AI/ML Model Development

Train predictive models for growth optimization, disease detection, and nutrient balancing.

IoT Platform Integration

Establish cloud-based platform for data ingestion, processing, and remote monitoring.

Automated Control Loop

Implement algorithms to adjust parameters based on AI predictions for self-optimization.

User Interface

Create web-based dashboard for real-time monitoring, control, and alerts.

Made with GAMMA

Data Description: The Lifeblood of Our AI

Sensor Data Streams

Continuous real-time measurements powering intelligent decisions:

- **pH:** Acidity/alkalinity (5.5-6.5 optimal)
- **EC:** Nutrient concentration (1.2-2.0 mS/cm)
- **Temperature:** Ambient and water (20-25°C)
- **Humidity:** Relative levels (60-75%)
- **Light:** PAR levels and LED spectrum

Image & Historical Data

Visual analysis and learning from experience:

- High-resolution images for growth tracking
- Pest and disease detection via computer vision
- Historical cultivation records for training

Made with GAMMA

Model/Approach: AI & IoT Synergy for Precision Growth

Core Technology

Internet of Things (IoT) for real-time data collection and remote control across all system components.

Predictive Growth Models

LSTM and Deep Neural Networks forecast optimal nutrient levels, light exposure, and environmental conditions for specific species.

Computer Vision Detection

Convolutional Neural Networks analyze plant imagery to detect early signs of pests, diseases, or nutrient deficiencies.

Nutrient Balancing

Reinforcement Learning and Regression automate precise dispensing of NPK and micronutrients based on sensor readings.

> **Closed-Loop Logic:** IoT sensors feed data to AI models, which trigger actuators to maintain ideal conditions, creating a self-optimizing environment.

Made with GAMMA

Justification of Method: Why AI and IoT?

Precision & Efficiency

AI algorithms analyze complex data patterns far beyond human capacity, enabling hyper-precise control of environmental factors for optimal growth.

Proactive Intervention

Predictive models and image analysis allow for early detection and mitigation of issues, preventing costly crop loss before it occurs.

Resource Optimization

Reduced water usage (up to 90% less than traditional), minimized pesticides, and optimized nutrients lead to cost savings and environmental benefits.

Scalability & Accessibility

Remote monitoring and automated control make cultivation feasible in diverse locations, including urban environments with limited space.

Data-Driven Insights

Continuous data collection provides valuable insights for refining cultivation protocols and improving future yields systematically.

Made with GAMMA

Results: Transforming Cultivation Outcomes

20-...

Faster Growth

Enhanced growth rates due to optimized conditions

50%

Reduced Losses

Decrease in crop loss from pests and diseases

90%

Water Savings

Reduction in water consumption vs. traditional

Enhanced Quality

Higher concentration of beneficial compounds, vibrant coloration, and superior morphological characteristics in exotic plants.

Remote Management

Full system monitoring and control from any internet-connected device, enabling efficient management and rapid response.

Made with GAMMA

Conclusion: The Future of Exotic Plant Cultivation is Smart

Sustainable Production

NutriGrow AI offers a sustainable, resource-efficient model for growing high-value exotic plants with minimal environmental impact.

Technological Integration

The synergy of IoT and AI unlocks unprecedented levels of control and optimization in controlled environment agriculture.

Market Potential

Enables consistent, high-quality production of rare plants, opening new markets and commercial opportunities.

Conservation Vision

To be the leading platform for intelligent cultivation of exotic flora, contributing to biodiversity conservation and sustainable agriculture.

Made with GAMMA

Questions?

Made with GAMMA

Thank You

NAME: TEJASHWINI.K

IDNO: AMB1259

Made with GAMMA